Bayesian Inference

In the Bayesian approach, the prior distribution is $T|\pi \sim Binom(n = 170, \pi)$.

The parameter of interest is $\psi = \frac{1-2\pi}{1-\pi}$

The initial step is to find a and b. The vaccine only meets the FDA guidelines if the vaccine efficacy is 30%. Therefore, we can use the probability:

$P(\psi \geq 0.3)$

Substituting ψ for $\frac{1-2\pi}{1-\pi}$

$P(\psi \geq 0.3) = P(\frac{1-2\pi}{1-\pi} \geq 0.3)$

Rearranging the equation $\frac{1-2\pi}{1-\pi} \geq 0.3$

$1 - 2\pi \geq 0.3(1 - \pi)$

$1 - 2\pi \geq 0.3 - 0.3\pi$

$1 \geq 0.3 + 1.7\pi$

$1 - 0.3 \geq 1.7\pi$

$0.7 \geq 1.7\pi$

$\frac{0.7}{1.7} \geq \pi$

$\pi \leq 0.4117$

Next, choosing our apriori beliefs. We chose to use strict parameters like:

$P(\psi \geq 0.3)0.05$ and $P(\psi \geq 0)0.5$.

$P(\psi \geq 0.3) = 0.05$ is that we believe there is only a 5% chance that the vaccine meets the FDA requirement of at least 30% efficacy.

$P(\psi \geq 0) = 0.5$ is that we believe there is a 50% chance that the vaccine is effective.

In terms of π, the beliefs are

$P(\psi \geq 0.3) = 0.05$ is $P(\pi \leq 0.4117) = 0.05$

$P(\psi \geq 0)0.5$ is $P(\pi \leq 0.5) = 0.5$

Next, to find a $Beta(a, b)$ prior for π, we used the library `LearnBayes` and the function `beta.select()`.

```
library(LearnBayes)
```

```
## Warning: package 'LearnBayes' was built under R version 4.4.3
```

```
beta.select(quantile1 = list(p = 0.05, x = 07/17),
            quantile2 = list(p = 0.5, x = 0.5))
```

```
## [1] 43.03 43.03
```

Using the function, our prior belief is $Beta(43.03, 43.03)$. To find the posterior distribution of π, we used this equation: $h(\pi|t) = Beta(43.03 + t, 170 - t + 43.03$ where $t = 8$.

The data model is $f(T|\pi) \sim Binom(170, \pi)$.

The prior is $g(\pi) = Beta(43.03, 43.03)$

The posterior is $h(\pi|t) = Beta(51.03, 205.03)$

This is the visualization of the posterior distributon of π

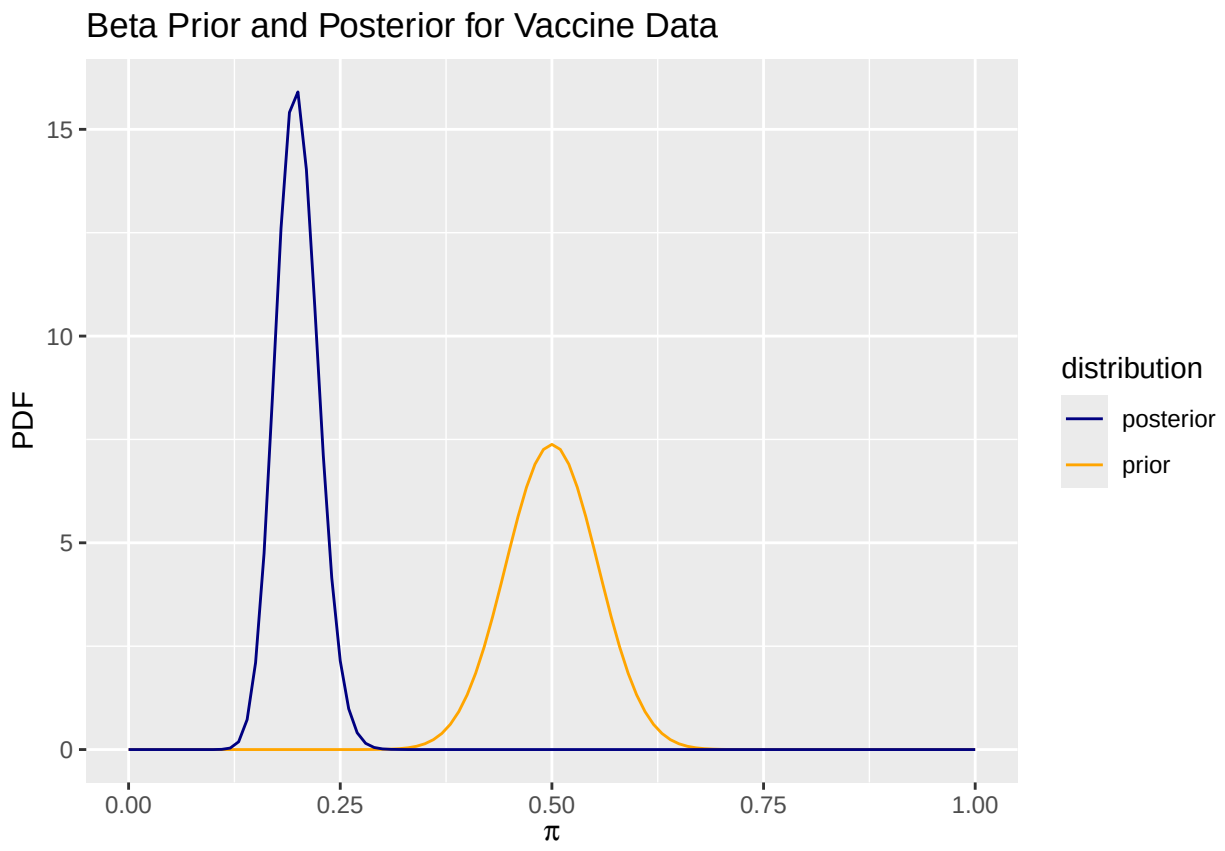
```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
ggplot()+
  stat_function(
    fun = dbeta, args = list(shape1 = 43.03, shape2 = 43.03),
    aes(color = "prior")) +
  stat_function(
    fun = dbeta,
    args = list(shape1 = 51.03, shape2 = 205.03),
```

```
    aes(color = "posterior")) +
  scale_color_manual(name = "distribution",
  values = c("prior" = "orange", "posterior" = "navyblue")) +

  labs(title = "Beta Prior and Posterior for Vaccine Data",
       x = expression(pi),
       y = "PDF")
```



Bayesian Analysis: Inference

#Posterior Distribution of pi

```
posterior_median_pi <- qbeta(0.5, shape1 = 43.03 + 8, shape2 = 170 - 8 + 43.03)

cat("The Posterior Median of pi is", round(posterior_median_pi, 4))
```

```
## The Posterior Median of pi is 0.1985
```

```
#95% Confidence Interval

cl <- qbeta(p = c(0.025, 0.975), shape1 = 43.03 + 8, shape2 = 170 - 8 + 43.03)

lower <- cl[1]

upper <- cl[2]

cat("The 95% Confidence Interval is from", lower, "to", upper)
```

```
## The 95% Confidence Interval is from 0.1527583 to 0.2502675
```

```
library(HDInterval)

hdi(qbeta, credMass = 0.95, shape1 = 43.03 + 8, shape2 = 170 - 8 + 43.03)
```

```
##     lower     upper
## 0.1512914 0.2486064
## attr(,"credMass")
## [1] 0.95
```

The hdi() function finds the shortest interval containing 95% of the posterior probability. The 95% highest posterior density interval for π is (0.1513, 0.2486).

Translating the median back to ψ

The Posterior Median of π is 0.1985 or $P(\pi < 0.1985|t) = 0.5$.

The parameter of interest, $\psi = \frac{1-2\pi}{1-\pi}$, can be rearranged in terms of π to $\pi = \frac{1-\psi}{2-\psi}$.

If $\pi = \frac{1-\psi}{2-\psi}$, then $P(\pi < 0.1985|t) = 0.5$ becomes

$P(\frac{1-\psi}{2-\psi} < 0.1985|t) = 0.5$

$P(1-\psi < 0.1985(2-\psi)|t) = 0.5$

$P(1-\psi < 0.397 - 0.1985\psi|t) = 0.5$

$P(-0.8015\psi < -0.603|t) = 0.5$

$P(\psi > 0.7523|t) = 0.5$

The posterior median of ψ is 0.7523.

Translating the 95% interval for π

The 95% confidence interval is from $0.1528 < \pi < 0.2503$.

To transform the interval for ψ, I'm using the parameter of interest, $\frac{1-2\pi}{1-\pi}$ and plugging in the π values.

For the π value: 0.1528

$\psi_{lower} = \frac{1-2(0.1528)}{1-0.1528} = \frac{0.6944}{0.8472} = 0.8196$

For the π value: 0.2503

$\psi_{upper} = \frac{1-2(0.2503)}{1-0.2503} = \frac{0.4994}{0.7497} = 0.6661$

The 95% confidence interval for π is (0.1528, 0.2503) while the interval for ψ is approximately (0.6661, 0.8196).

The posterior distribution suggests the vaccine efficacy is likely betwen 66.61% to 81.96%.

Bayesian P-value

The null and alternative hypothesis are

$H_0 : \psi = 0.3$ vs $H_1 : \psi > 0.3$

To find the the posterior probability $P(\psi \leq 0.3|t)$, I'll use the parameter of interest.

The parameter of interest, $\psi = \frac{1-2\pi}{1-\pi}$, can be rearranged in terms of π to $\pi = \frac{1-\psi}{2-\psi}$.

I'll plug in $\psi = 0.3$ into $\pi = \frac{1-\psi}{2-\psi}$ to get:

$\pi = \frac{1-0.3}{2-0.3} = \frac{0.7}{1.7} = 0.4118$

$\pi = 0.4118$

To compute the Bayesian p-value, we use the `pbeta` function and $\pi \geq 0.4118$.

$P(\pi \geq 0.4118|t)$

```
pi <- 0.7/1.7

a <- 51.03

b <- 205.03

p_val <- 1 - pbeta(pi, a, b)

p_val
```

```
## [1] 1.931788e-13
```

The p-value is $1.9317881 \times 10^{-13}$ which is extremely small indicating we reject the null hypothesis and side with the alternative that the vaccine efficacy exceeds the FDA threshold of 30%.

Results

Present your findings.

Bayesian Method

Through Bayesian Inference, we denote T as the number of COVID-19 cases in the vaccine group modeled as $T|\pi \sim Binomial(n = 170, \pi)$. The parameter of interest is vaccine efficacy which is modeled with $\psi = \frac{1-2\pi}{1-\pi}$

The prior beliefs we have is that the there is a 5% prior probability that the vaccine meets the FDA efficacy requirement of at least 30% ($P(\psi \geq 0.3) = 0.05$).

With our calculations, we know that 0.4117 is the cutoff infection probability that corresponds to the FDA's 30% vaccine efficacy.

Based on the data and prior assumptions, the Posterior Median of π is 0.1985. This means that the median estimate of the true infection probability among vaccinated individuals is 19.85%.

The 95% confidence interval for π is [0.1528, 0.2503] or that the true probability of infection among vaccinated individuals is likely between 15.28% to 25.03%.

The 95% highest posterior density interval for π is (0.1513, 0.2486). There is a 95% posterior probability that the true infection probability among vaccinated individuals is likely between 15.13% to 24.86%.

The posterior median of *psi* is 0.7523. This means that the vaccine is estimated to reduce the probability of infection by approximately 75.23%.

The 95% confidence interval for *psi* is [0.6661, 0.8196]. There is a 95% posterior probability that the true vaccine efficacy lies between 66.61% to 81.96%.

With the null and alternative hypothesis being $H_0 : \psi = 0.3$ vs $H_1 : \psi > 0.3$ for vaccine efficacy, the p-value (1.9317881$\times$ 10$-$13) is smaller than the significance levels of 0.05 and 0.01 indicating that the vaccine efficacy exceeds the FDA threshold of 30%.